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November 11, 2025, 2:40 PM

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Document Type:	Journal	

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Reactions (50)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%

Multi-component
structure image
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SciFinder



Multi-component
structure image
available in CAS
SciFinder

 Supplier (1)

31-080-CAS-7753712

Steps: 1 Yield: 100%

- 1.1 **Catalysts:** Rhodium, bis[μ-[α,α',α'',α''']-tetramethyl-1,3-benzenedipropionato(2-)-κO¹,κO³:κO³,κO¹]]di-, (*Rh-Rh*)
Solvents: 1,2-Dichloroethane; 15 h, 80 °C

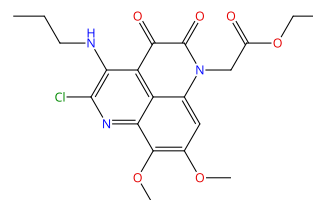
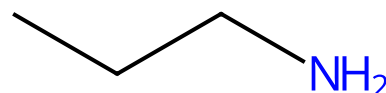
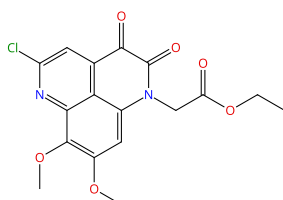
Rh^{II}-Catalyzed Synthesis of α-, β-, or δ-Carbolines from Aryl Azides

By: Pumphrey, Ashley L.; et al

Angewandte Chemie, International Edition (2012), 51(24), 5920-5923, S5920/1-S5920/204.

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Supplier (1)

 Suppliers (69)

31-080-CAS-4032036

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Triethylamine
Solvents: Tetrahydrofuran; 0.5 h, rt

Novel Amination and 1,2-Amino,hydro-Elimination between 2,3-Diketopyrido[4,3,2-de]quinolines and Primary Amino Compounds

By: Ding, Qizhu; et al

Journal of Organic Chemistry (1999), 64(21), 7965-7972.

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%

Multi-component
structure image
available in CAS
SciFinder



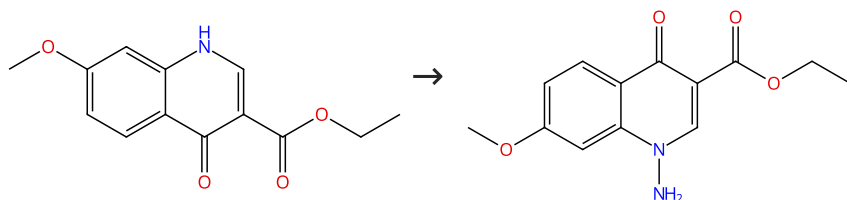
Multi-component
structure image
available in CAS
SciFinder

 Supplier (1)

31-080-CAS-11549435 Steps: 1 Yield: 100% 1.1 Catalysts: Rhodium, bis[μ-[α,α,α',α'-tetramethyl-1,3-benzenedipropanoato(2-)-κO¹,κO³:κO³,κO¹]]di-, (<i>Rh-Rh</i>) Solvents: 1,2-Dichloroethane; 15 h, 80 °C	Rh^{II}₂-Catalyzed Synthesis of α-, β-, or δ-Carbolines from Aryl Azides By: Pumphrey, Ashley L.; et al Angewandte Chemie, International Edition (2012), 51(24), 5920-5923, S5920/1-S5920/204.
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Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (47)

31-080-CAS-9521923 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate, Benzenesulfonic acid, 2,4,6-trimethyl-, azanyl ester, 2,2,2-trifluoroacetate (1:1) Solvents: Dimethylformamide 1.2 Solvents: Water Experimental Protocols	A convenient procedure for N-amination of 4-oxo-1,4-dihydroquinolines By: Edmont, Dolores; et al Synthetic Communications (2000), 30(2), 217-225.
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Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

Suppliers (3)

31-080-CAS-14757414 Steps: 1 Yield: 100% 1.1 Catalysts: Rhodium, bis[μ-[α,α,α',α'-tetramethyl-1,3-benzenedipropanoato(2-)-κO¹,κO³:κO³,κO¹]]di-, (<i>Rh-Rh</i>) Solvents: 1,2-Dichloroethane; 15 h, 80 °C	Rh^{II}₂-Catalyzed Synthesis of α-, β-, or δ-Carbolines from Aryl Azides By: Pumphrey, Ashley L.; et al Angewandte Chemie, International Edition (2012), 51(24), 5920-5923, S5920/1-S5920/204.
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Scheme 6 (1 Reaction)

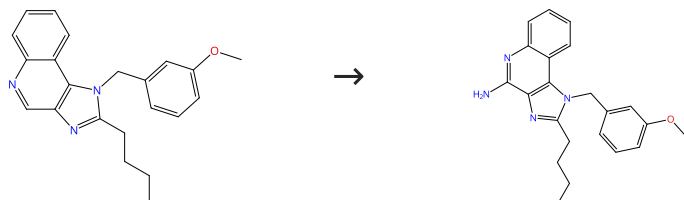
Steps: 1 Yield: 100%



31-080-CAS-3820235	Steps: 1 Yield: 100%	Rh^{II}₂-Catalyzed Synthesis of α-, β-, or δ-Carbolines from Aryl Azides
1.1 Catalysts: Rhodium, bis[μ -[$\alpha,\alpha,\alpha',\alpha'$ -tetramethyl-1,3-benzenedipropanoato(2-)- $\kappa O^1,\kappa O^3:\kappa O^3,\kappa O^1$]]di-, (<i>Rh-Rh</i>) Solvents: 1,2-Dichloroethane; 15 h, 80 °C		By: Pumphrey, Ashley L.; et al Angewandte Chemie, International Edition (2012), 51(24), 5920-5923, S5920/1-S5920/204.

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-37912551	Steps: 1 Yield: 100%	Synthesis and Optimization of 1-Substituted Imidazo[4,5-c]quinoline TLR7 Agonists
1.1 Reagents: <i>m</i> -Chloroperbenzoic acid Solvents: Chloroform; 1 h, 40 °C; 40 °C → 50 °C		By: DeYoung, Emma G.; et al
1.2 Reagents: Tosyl chloride, Ammonium hydroxide; 50 °C; 50 °C → rt; 1 h, rt		ACS Medicinal Chemistry Letters (2023), 14(10), 1358-1368.
Experimental Protocols		

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



31-080-CAS-9880497	Steps: 1 Yield: 100%	Rh^{II}₂-Catalyzed Synthesis of α-, β-, or δ-Carbolines from Aryl Azides
1.1 Catalysts: Rhodium, bis[μ -[$\alpha,\alpha,\alpha',\alpha'$ -tetramethyl-1,3-benzenedipropanoato(2-)- $\kappa O^1,\kappa O^3:\kappa O^3,\kappa O^1$]]di-, (<i>Rh-Rh</i>) Solvents: 1,2-Dichloroethane; 15 h, 80 °C		By: Pumphrey, Ashley L.; et al Angewandte Chemie, International Edition (2012), 51(24), 5920-5923, S5920/1-S5920/204.

Scheme 9 (1 Reaction)

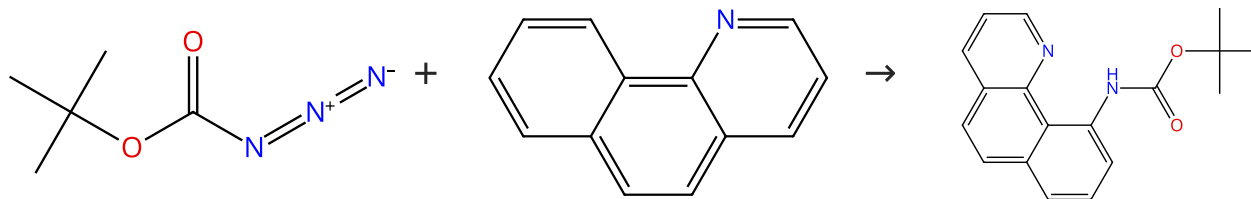
Steps: 1 Yield: 100%



31-080-CAS-14137566	Steps: 1 Yield: 100%	Rh^{II}₂-Catalyzed Synthesis of α-, β-, or δ-Carbolines from Aryl Azides
1.1 Catalysts: Rhodium, bis[μ -[$\alpha,\alpha,\alpha',\alpha'$ -tetramethyl-1,3-benzenedipropanoato(2-)- $\kappa O^1,\kappa O^3:\kappa O^3,\kappa O^1$]]di-, (<i>Rh-Rh</i>) Solvents: 1,2-Dichloroethane; 15 h, 80 °C		By: Pumphrey, Ashley L.; et al Angewandte Chemie, International Edition (2012), 51(24), 5920-5923, S5920/1-S5920/204.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (12)

Suppliers (72)

31-080-CAS-15783535

Steps: 1 Yield: 100%

Rhodium-Catalyzed N-tert-Butoxycarbonyl (Boc) Amination by Directed C-H Bond Activation

By: Wippich, Julian; et al

Advanced Synthesis & Catalysis (2016), 358(13), 2083-2087.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Bis[dichloro[η⁵-(pentamethylcyclopentadienyl)]rhodium], Silver hexafluoroantimonate
Solvents: 1,2-Dichloroethane; 14 h, 80 °C

Experimental Protocols

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



31-080-CAS-1411064

Steps: 1 Yield: 100%

Rh^{II}₂-Catalyzed Synthesis of α-, β-, or δ-Carbolines from Aryl Azides

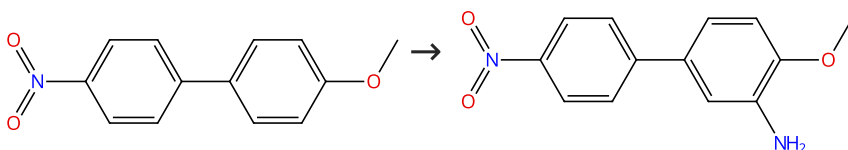
By: Pumphrey, Ashley L.; et al

Angewandte Chemie, International Edition (2012), 51(24), 5920-5923, S5920/1-S5920/204.

1.1 **Catalysts:** Rhodium, bis[μ-[α,α,α',α'-tetramethyl-1,3-benzenedipropanoato(2-)-κO¹,κO³:κO³,κO¹]]di-, (*Rh-Rh*)
Solvents: 1,2-Dichloroethane; 15 h, 80 °C

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (53)

Supplier (1)

31-080-CAS-2062741

Steps: 1 Yield: 100%

Electrochemical C-H Amination: Synthesis of Aromatic Primary Amines via N-Arylpyridinium Ions

By: Morofuji, Tatsuya; et al

Journal of the American Chemical Society (2013), 135(13), 5000-5003.

1.1 **Reagents:** Pyridine, Tetrabutylammonium tetrafluoroborate, Trifluoromethanesulfonic acid
Solvents: Acetonitrile; 25 °C
 1.2 **Reagents:** Piperidine; 12 h, 80 °C

Experimental Protocols

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%

Multi-component
structure image
available in CAS
SciFinder



Multi-component
structure image
available in CAS
SciFinder

31-080-CAS-10159923

Steps: 1 Yield: 100%

1.1 **Catalysts:** Rhodium, bis[μ-[α,α,α',α'-tetramethyl-1,3-benzenedipropanoato(2-)-κO¹,κO³:κO³,κO¹]]di-, (*Rh-Rh*)
Solvents: 1,2-Dichloroethane; 15 h, 80 °C

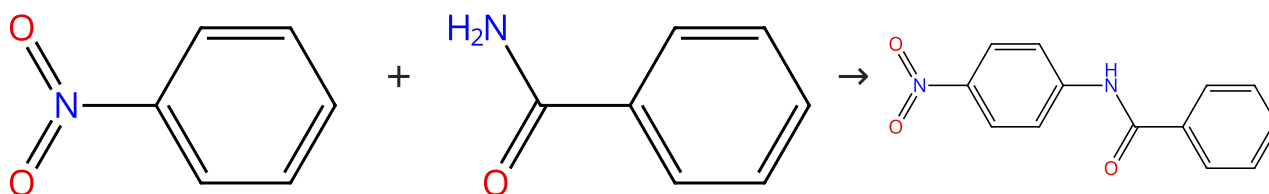
Rh^{II}₂-Catalyzed Synthesis of α-, β-, or δ-Carbolines from Aryl Azides

By: Pumphrey, Ashley L.; et al

Angewandte Chemie, International Edition (2012), 51(24), 5920-5923, S5920/1-S5920/204.

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (103)

Suppliers (106)

Suppliers (65)

31-080-CAS-11630668

Steps: 1 Yield: 100%

No Data Available

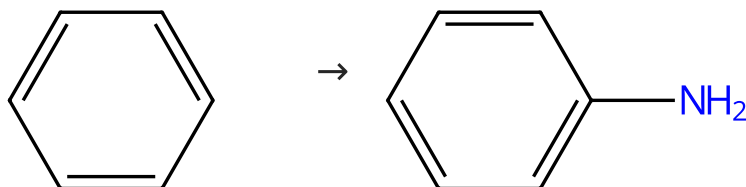
Amination of nitrobenzene via nucleophilic aromatic substitution for hydrogen: direct formation of aromatic amide bonds

By: Stern, Michael K.; et al

Journal of Organic Chemistry (1993), 58(24), 6883-8.

Scheme 15 (3 Reactions)

Steps: 1 Yield: 100%



Suppliers (174)

Suppliers (113)

31-080-CAS-23646992

Steps: 1 Yield: 100%

1.1 **Reagents:** Acetic acid, Sulfuric acid, Hydroxylamine
Catalysts: Manganese oxide (MnO₂), Graphene (oxide, reduced); 30 min, rt

1.2 90 min, 70 °C

Experimental Protocols

Manganese oxide supported partially reduced graphene oxide as a highly active and durable catalyst for the amination of benzene

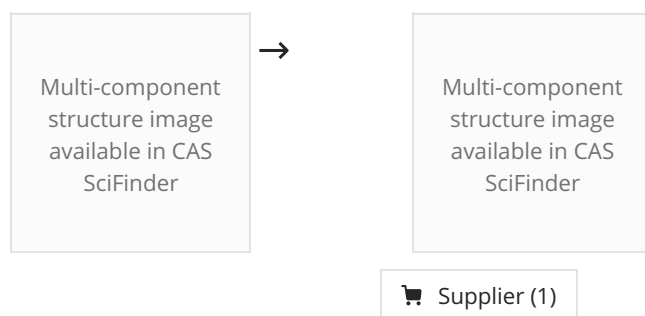
By: Rana, Surjakanta; et al

Catalysis Communications (2021), 157, 106329.

31-080-CAS-6516011 Steps: 1 Yield: 100% 1.1 Reagents: Hydrazoic acid Catalysts: Trifluoroacetic acid, Trifluoromethanesulfonic acid Solvents: Chloroform	Direct aromatic amination by azides: reactions of hydrazoic acid and butyl azides with aromatic compounds in the presence of both trifluoromethanesulfonic acid and trifluoroacetic acid By: Takeuchi, Hiroshi; et al Journal of the Chemical Society, Perkin Transactions 1: Organic and Bio-Organic Chemistry (1972-1999) (1993), (7), 867-70.
31-080-CAS-4655445 Steps: 1 Yield: 100% 1.1 Reagents: Trifluoroacetic acid, Trifluoromethanesulfonic acid, Hydrazoic acid Solvents: Chloroform	Efficient direct aromatic amination by hydrazoic acid in the presence of both trifluoromethanesulfonic acid and trifluoroacetic acid By: Takeuchi, Hiroshi; et al Journal of the Chemical Society, Chemical Communications (1991), (21), 1524-5.

Scheme 16 (1 Reaction)

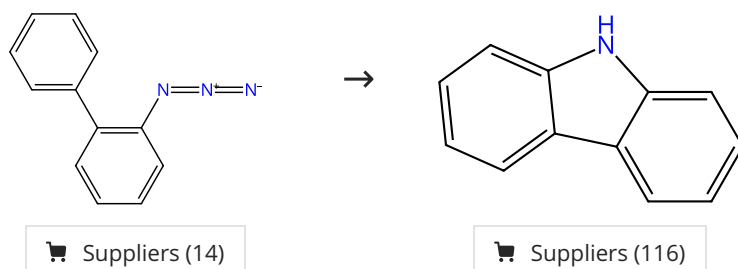
Steps: 1 Yield: 100%



31-080-CAS-8032996 Steps: 1 Yield: 100% 1.1 Catalysts: Rhodium, bis[μ-[α,α,α',α'-tetramethyl-1,3-benzenedipropanoato(2-)-κO¹,κO³:κO³,κO¹]]di-, (<i>Rh-Rh</i>) Solvents: 1,2-Dichloroethane; 15 h, 80 °C	Rh^{II}₂-Catalyzed Synthesis of α-, β-, or δ-Carbolines from Aryl Azides By: Pumphrey, Ashley L.; et al Angewandte Chemie, International Edition (2012), 51(24), 5920-5923, S5920/1-S5920/204.
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Scheme 17 (1 Reaction)

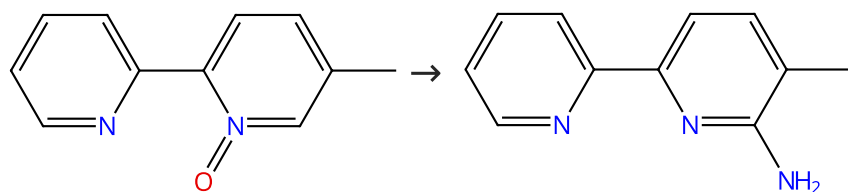
Steps: 1 Yield: 100%



31-080-CAS-21050048 Steps: 1 Yield: 100% 1.1 Reagents: Silica Solvents: Acetone, Water; 36 h, 28 - 32 °C Experimental Protocols	Visible-light-promoted intramolecular C-H amination in aqueous solution: synthesis of carbazoles By: Yang, Lizheng; et al Green Chemistry (2018), 20(6), 1362-1366.
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Scheme 18 (1 Reaction)

Steps: 1 Yield: 99%



Supplier (1)

31-614-CAS-40025616

Steps: 1 Yield: 99%

Simple and Efficient Method for Mono- and Di-Amination of Polypyridine N-Oxides

By: Verbeet, Wessel; et al

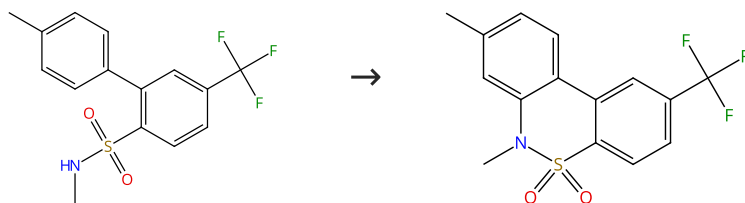
European Journal of Organic Chemistry (2024), 27(14), e202400054.

1.1 **Reagents:** Tosyl chloride, Triethylamine, Potassium phthalimide**Solvents:** Dichloromethane; 0 °C; 24 h, rt1.2 **Reagents:** Hydrazine hydrate (1:1)**Solvents:** Water; 6 h, 80 °C

Experimental Protocols

Scheme 19 (1 Reaction)

Steps: 1 Yield: 99%



31-080-CAS-18162222

Steps: 1 Yield: 99%

Synthesis of benzosultams via an intramolecular sp^2 C-H bond amination reaction of o-arylbenzenesulfonamides under metal-free conditions

By: Li, Yuewen; et al

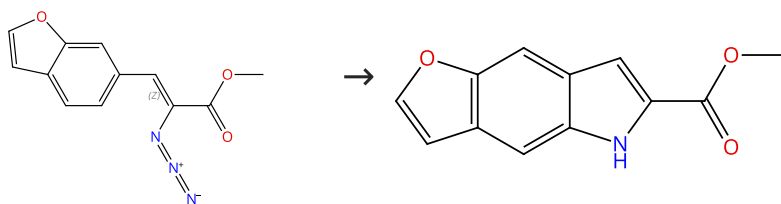
Organic & Biomolecular Chemistry (2014), 12(1), 149-155.

1.1 **Reagents:** Potassium carbonate, Iodobenzene diacetate, Iodine**Solvents:** Dichloromethane; 2 h, 35 °C

Experimental Protocols

Scheme 20 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Supplier (1)

Suppliers (2)

31-080-CAS-2735031

Steps: 1 Yield: 99%

Iron(II) Triflate as a Catalyst for the Synthesis of Indoles by Intramolecular C-H Amination

By: Bonnamour, Julien; et al

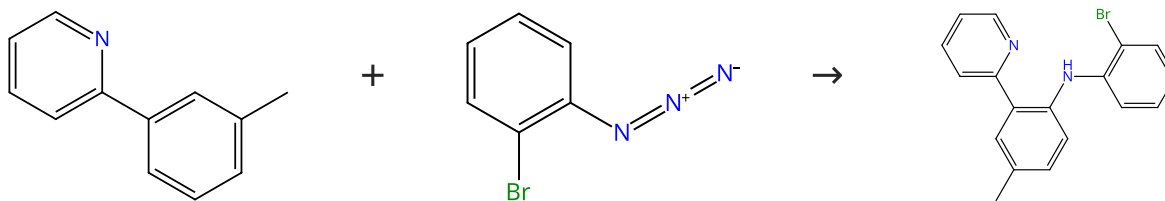
Organic Letters (2011), 13(8), 2012-2014.

1.1 **Catalysts:** Methanesulfonic acid, 1,1,1-trifluoro-, iron(2+) salt (2:1)**Solvents:** Tetrahydrofuran; 24 h, 80 °C

Experimental Protocols

Scheme 21 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (56)

Suppliers (18)

31-080-CAS-11103079

Steps: 1 Yield: 99%

Rhodium-Catalyzed Selective Mono- and Diamination of Arenes with Single Directing Site "On Water"

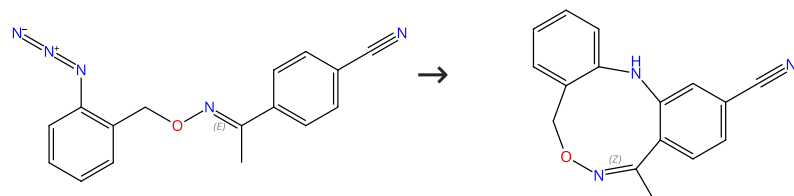
- 1.1 **Catalysts:** Bis[dichloro[η⁵-(pentamethylcyclopentadienyl)]rhodium], Sodium tetrakis[3,5-bis(trifluoromethyl)phenyl] borate
Solvents: Water; 24 h, 110 °C; 110 °C → rt

By: Ali, Ashif Md; et al

Organic Letters (2016), 18(6), 1386-1389.

Scheme 22 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Double bond geometry shown

31-080-CAS-16830470

Steps: 1 Yield: 99%

Intramolecular Amido Transfer Leading to Structurally Diverse Nitrogen-Containing Macrocycles

- 1.1 **Catalysts:** Sodium acetate, Bis[dichloro[η⁵-(pentamethylcyclopentadienyl)]rhodium], [1,1,1-Trifluoro-*N*-[(trifluoromethyl)sulfonyl-κ*O*]methanesulfonamidato-κ*O*]silver
Solvents: 1,2-Dichloroethane; 12 h, 60 °C

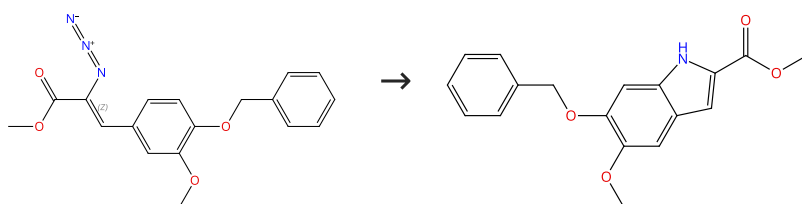
By: Kim, Heejeong; et al

Angewandte Chemie, International Edition (2017), 56(12), 3344-3348.

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (10)

Supplier (1)

31-080-CAS-13982596

Steps: 1 Yield: 99%

Iron(II) Triflate as a Catalyst for the Synthesis of Indoles by Intramolecular C-H Amination

- 1.1 **Catalysts:** Methanesulfonic acid, 1,1,1-trifluoro-, iron(2+) salt (2:1)
Solvents: Tetrahydrofuran; 24 h, 80 °C

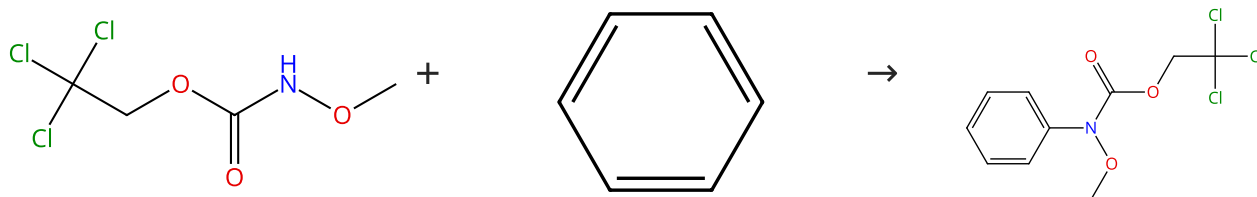
By: Bonnamour, Julien; et al

Organic Letters (2011), 13(8), 2012-2014.

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 99%



Supplier (1)

Suppliers (174)

31-080-CAS-21172936

Steps: 1 Yield: 99%

An Improved Catalyst for Iodine(I/III)-Catalysed Intermolecular CXH Amination

By: Lucchetti, Nicola; et al

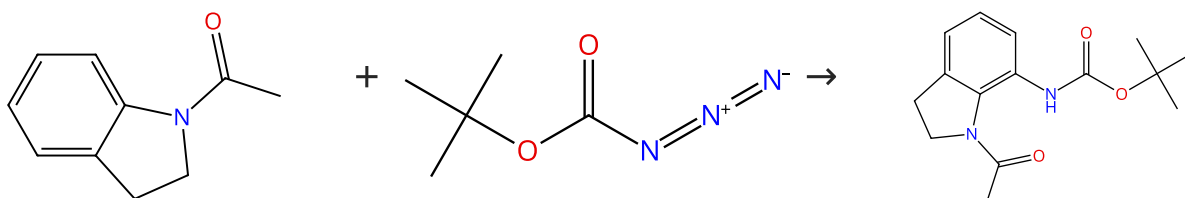
Advanced Synthesis & Catalysis (2016), 358(13), 2093-2099.

1.1 **Reagents:** Trifluoroacetic acid, 1,3-Bis(acetyloxy)-1,3-dihydro-1,3,2-benzodioxole
Solvents: 1,2-Dichloroethane, 1,1,1,3,3,3-Hexafluoro-2-propanol; 2 h, 25 °C

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (75)

Suppliers (12)

Supplier (1)

31-080-CAS-15759075

Steps: 1 Yield: 99%

A Facile Access to Primary Alkylamines and Anilines via Ir(III)-Catalyzed C-H Amination Using Azidoformates

By: Kim, Hyunwoo; et al

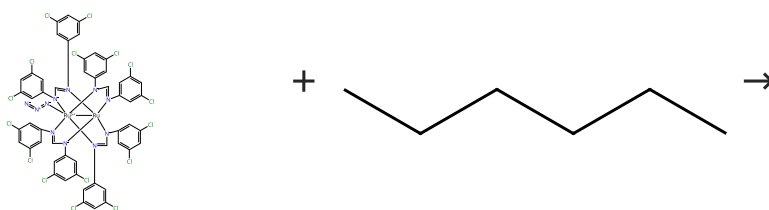
ACS Catalysis (2016), 6(9), 5922-5929.

1.1 **Reagents:** Sodium acetate, Oxygen
Catalysts: Di-μ-chlorodichlorobis(η⁵-2,4-cyclopentadien-1-yl)diiridium, [1,1,1-Trifluoro-*N*-[(trifluoromethyl)sulfonyl-κ*O*]methanesulfonamidato-κ*O*]silver
Solvents: 1,2-Dichloroethane; 24 h, 60 °C

Experimental Protocols

Scheme 26 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (334)

Multi-component
structure image
available in CAS
SciFinder

31-080-CAS-9176276

Steps: 1 Yield: 99%

Aryl C-H Amination by Diruthenium Nitrides in the Solid State and in Solution at Room Temperature: Experimental and Computational Study of the Reaction Mechanism

By: Long, Amanda Kae Musch; et al

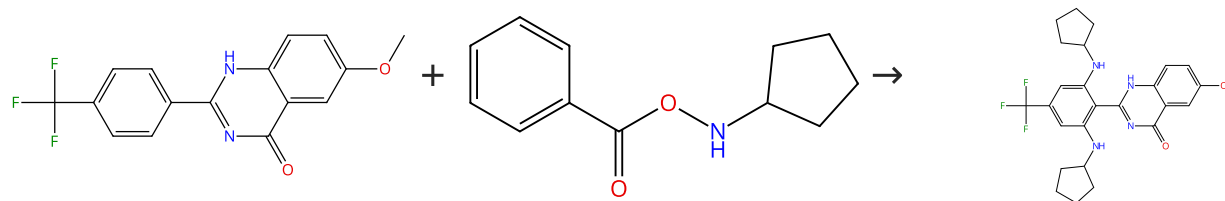
Journal of the American Chemical Society (2011), 133(33), 13138-13150.

1.1 48 h, 160 °C
 1.2 **Solvents:** Dichloromethane; rt

Experimental Protocols

Scheme 27 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (6)

31-080-CAS-19248524

Steps: 1 Yield: 99%

Rhodium(III)-catalyzed C-H amination of 2-arylquinazolin-4(3H)-one with N-alkyl-O-benzoyl-hydroxylamines

By: Zhang, Yuanguang; et al

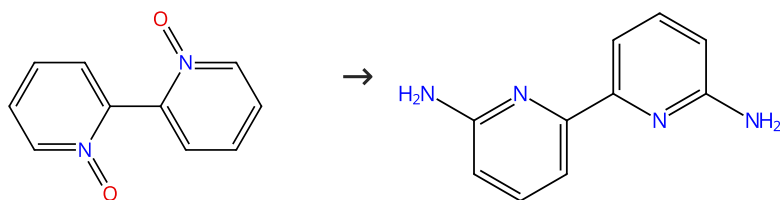
Tetrahedron (2018), 74(19), 2330-2337.

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: Pivalic acid, Bis[dichloro[η⁵-(pentamethylcyclopentadienyl)]rhodium]
Solvents: Toluene; 12 h, 50 °C

Experimental Protocols

Scheme 28 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (73)

Suppliers (66)

31-614-CAS-40025625

Steps: 1 Yield: 99%

Simple and Efficient Method for Mono- and Di-Amination of Polypyridine N-Oxides

By: Verbeet, Wessel; et al

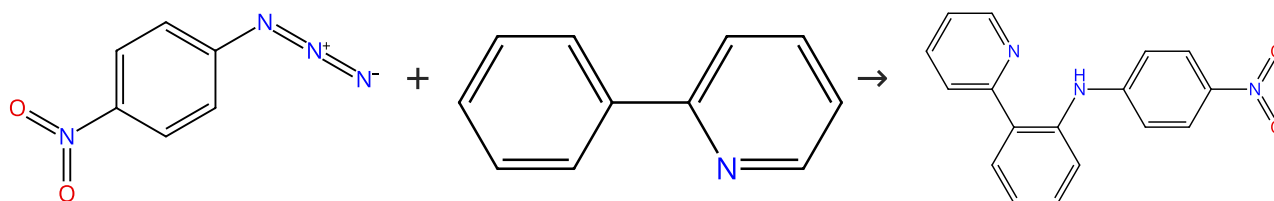
European Journal of Organic Chemistry (2024), 27(14), e202400054.

- 1.1 **Reagents:** Tosyl chloride, Triethylamine, Potassium phthalimide
Solvents: Acetonitrile; 24 h, rt
- 1.2 **Reagents:** Hydrazine hydrate (1:1)
Solvents: Water; 24 h, 80 °C

Experimental Protocols

Scheme 29 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (26)

Suppliers (90)

Supplier (1)

31-080-CAS-13884375

Steps: 1 Yield: 99%

Rhodium-Catalyzed Selective Mono- and Diamination of Arenes with Single Directing Site "On Water"

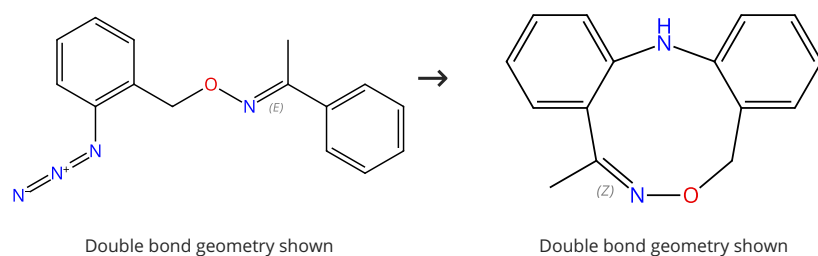
By: Ali, Ashif Md; et al

Organic Letters (2016), 18(6), 1386-1389.

- 1.1 **Catalysts:** Bis[dichloro[η⁵-(pentamethylcyclopentadienyl)]rhodium], Sodium tetrakis[3,5-bis(trifluoromethyl)phenyl]borate
Solvents: Water; 24 h, 110 °C; 110 °C → rt

Scheme 30 (1 Reaction)

Steps: 1 Yield: 99%



31-080-CAS-16830463

Steps: 1 Yield: 99%

Intramolecular Amido Transfer Leading to Structurally Diverse Nitrogen-Containing Macrocycles

By: Kim, Heejeong; et al

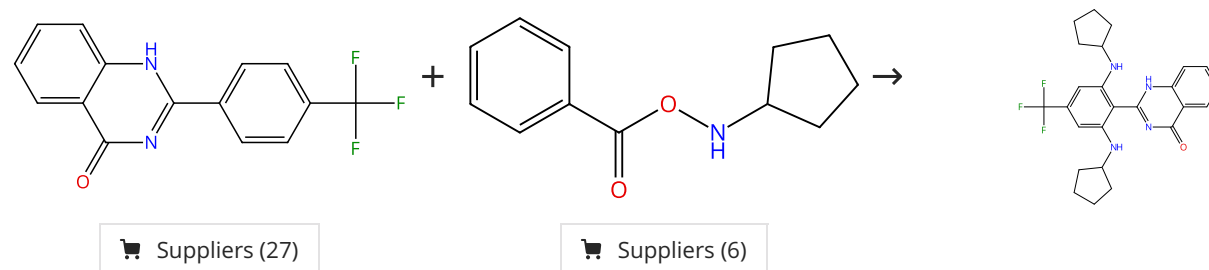
Angewandte Chemie, International Edition (2017), 56(12), 3344-3348.

1.1 **Catalysts:** Sodium acetate, Bis[dichloro[η⁵-(pentamethylcyclopentadienyl)]rhodium], [1,1,1-Trifluoro-*N*-[(trifluoromethyl)sulfonyl-κ O]methanesulfonamido-κ O]silver
Solvents: 1,2-Dichloroethane; 12 h, 60 °C

Experimental Protocols

Scheme 31 (1 Reaction)

Steps: 1 Yield: 99%



31-080-CAS-19248517

Steps: 1 Yield: 99%

Rhodium(III)-catalyzed C-H amination of 2-arylquinazolin-4(3H)-one with N-alkyl-O-benzoyl-hydroxylamines

By: Zhang, Yuanguang; et al

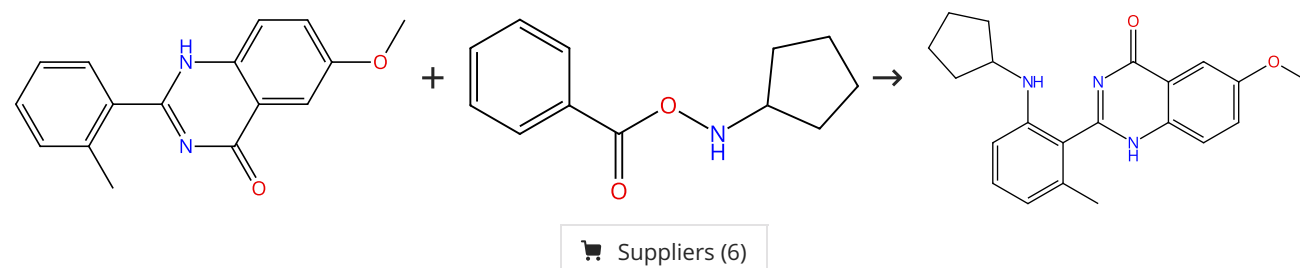
Tetrahedron (2018), 74(19), 2330-2337.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Pivalic acid, Bis[dichloro[η⁵-(pentamethylcyclopentadienyl)]rhodium]
Solvents: Toluene; 12 h, 50 °C

Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 99%



31-080-CAS-19248529

Steps: 1 Yield: 99%

Rhodium(III)-catalyzed C-H amination of 2-arylquinazolin-4(3H)-one with N-alkyl-O-benzoyl-hydroxylamines

By: Zhang, Yuanguang; et al

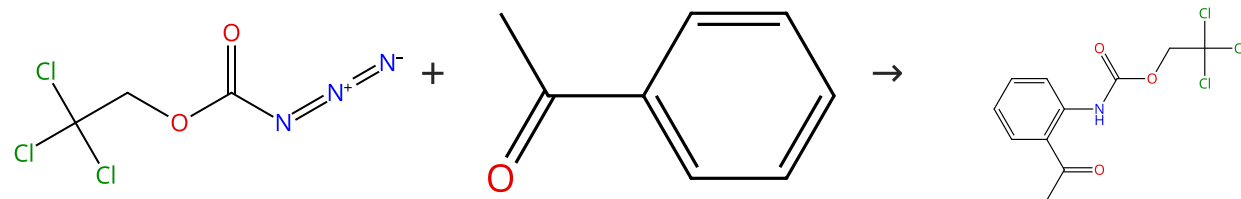
Tetrahedron (2018), 74(19), 2330-2337.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Pivalic acid, Bis[dichloro[η⁵-(pentamethylcyclopentadienyl)]rhodium]
Solvents: Toluene; 12 h, 50 °C

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (2)

Suppliers (105)

31-614-CAS-34343059

Steps: 1 Yield: 99%

Carbonyl-Assisted Iridium-Catalyzed C-H Amination Using 2,2,2-Trichloroethoxycarbonyl Azide

- 1.1 **Reagents:** Potassium acetate
Catalysts: Silver triflate, Iridium, di-μ-chlorodichlorobis[(1,2,3,4,5-η)-1,2,3,4,5-pentamethyl-2,4-cyclopentadien-1-yl]di-
Solvents: 1,2-Dichloroethane; 24 h, 80 °C

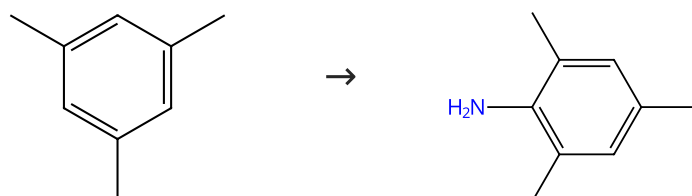
By: Dong, Xunqing; et al

Journal of Organic Chemistry (2022), 87(21), 13990-14004.

Experimental Protocols

Scheme 34 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (90)

Suppliers (73)

31-080-CAS-13169009

Steps: 1 Yield: 99%

Efficient direct aromatic amination by hydrazoic acid in the presence of both trifluoromethanesulfonic acid and trifluoroacetic acid

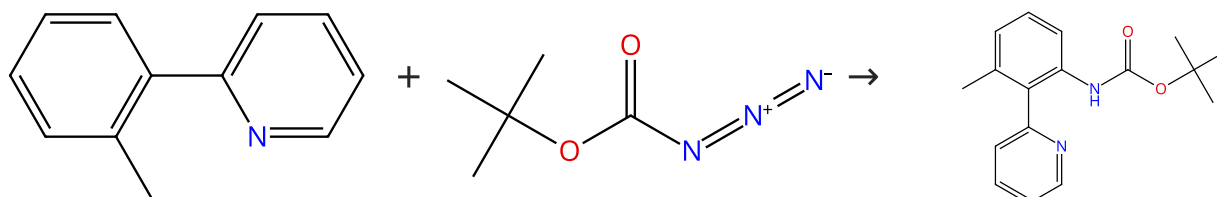
- 1.1 **Reagents:** Trifluoroacetic acid, Trifluoromethanesulfonic acid, Hydrazoic acid
Solvents: Chloroform

By: Takeuchi, Hiroshi; et al

Journal of the Chemical Society, Chemical Communications (1991), (21), 1524-5.

Scheme 35 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (64)

Suppliers (12)

Supplier (1)

31-080-CAS-15759072

Steps: 1 Yield: 99%

A Facile Access to Primary Alkylamines and Anilines via Ir(III)-Catalyzed C-H Amination Using Azidoformates

- 1.1 **Reagents:** Sodium acetate, Oxygen
Catalysts: Di-μ-chlorodichlorobis(η⁵-2,4-cyclopentadien-1-yl)diiridium, [1,1,1-Trifluoro-*N*-[(trifluoromethyl)sulfonyl-κ*O*]methanesulfonamidato-κ*O*]silver
Solvents: 1,2-Dichloroethane; 24 h, 60 °C

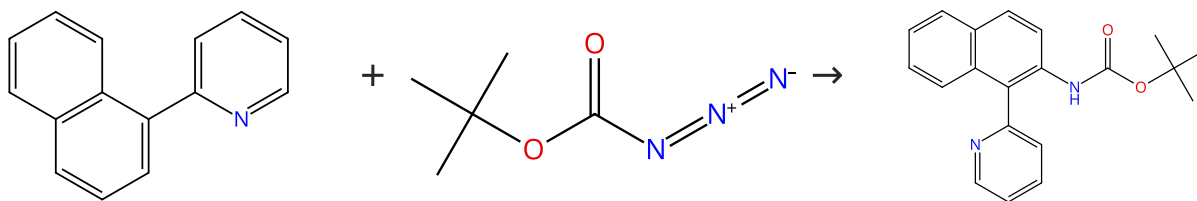
By: Kim, Hyunwoo; et al

ACS Catalysis (2016), 6(9), 5922-5929.

Experimental Protocols

Scheme 36 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (25)

Suppliers (12)

31-080-CAS-15759073

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Sodium acetate, Oxygen
Catalysts: Di- μ -chlorodichlorobis(η^5 -2,4-cyclopentadien-1-yl)diiridium, [1,1,1-Trifluoro-*N*-[(trifluoromethyl)sulfonyl- κO]methanesulfonamido- κO]silver
Solvents: 1,2-Dichloroethane; 24 h, 60 °C

Experimental Protocols

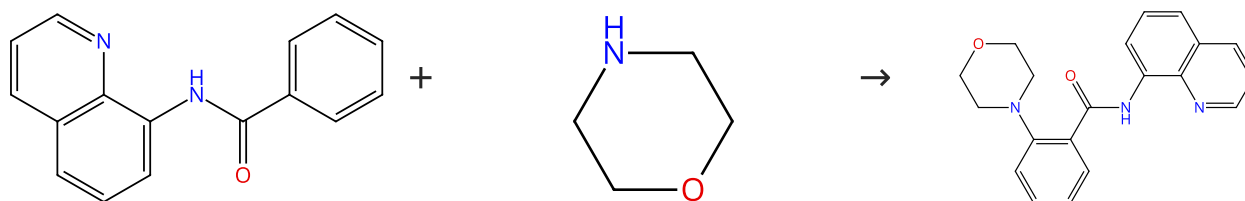
A Facile Access to Primary Alkylamines and Anilines via Ir(III)-Catalyzed C-H Amination Using Azidoformates

By: Kim, Hyunwoo; et al

ACS Catalysis (2016), 6(9), 5922-5929.

Scheme 37 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (27)

Suppliers (88)

Suppliers (2)

31-080-CAS-18139151

Steps: 1 Yield: 99%

- 1.1 **Reagents:** N-methylmorpholine N-oxide
Catalysts: Silver acetate, Triqua[μ -[1,3,5-benzenetricarboxylato(3-)- κO^1 : κO^1]][μ_3 -[1,3,5-benzenetricarboxylato(3-)- κO^1 : κO^3 : κO^1]]tricopper
Solvents: *N*-Methyl-2-pyrrolidone; 6 h, 90 °C
- 1.2 **Reagents:** Potassium hydroxide
Solvents: Water

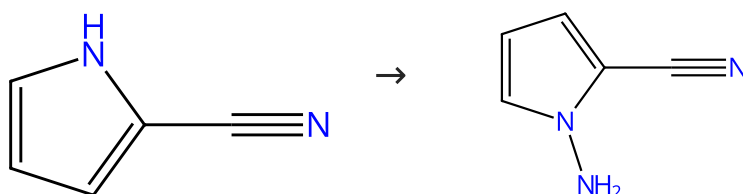
Removable bidentate directing group assisted-recyclable metal-organic frameworks-catalyzed direct oxidative amination of sp^2 C-H bonds

By: Tran, Nga T. T.; et al

Journal of Catalysis (2014), 320, 9-15.

Scheme 38 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (73)

Suppliers (54)

31-614-CAS-37644319

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Potassium *tert*-butoxide, Monochloramine
Solvents: Tetrahydrofuran, *tert*-Butyl methyl ether; 2 min, 60 °C

Experimental Protocols

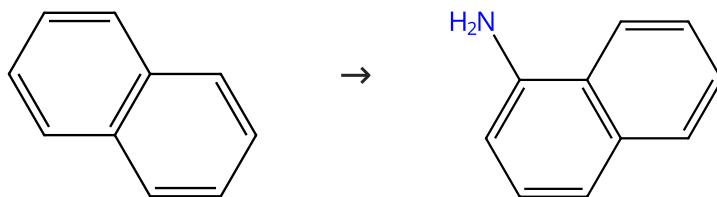
Development of Continuous Flow Processes to Access Pyrrolo [2,1-*f*] [1,2,4]triazin-4-amine: An RSM for the Synthesis of Antiviral Drugs

By: Vasilev, Maksim; et al

Organic Process Research & Development (2024), 28(5), 1556-1566.

Scheme 39 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (134)

Suppliers (66)

31-080-CAS-6095797

Steps: 1 Yield: 99%

Electrochemical C-H Amination: Synthesis of Aromatic Primary Amines via N-Arylpyridinium Ions

By: Morofuji, Tatsuya; et al

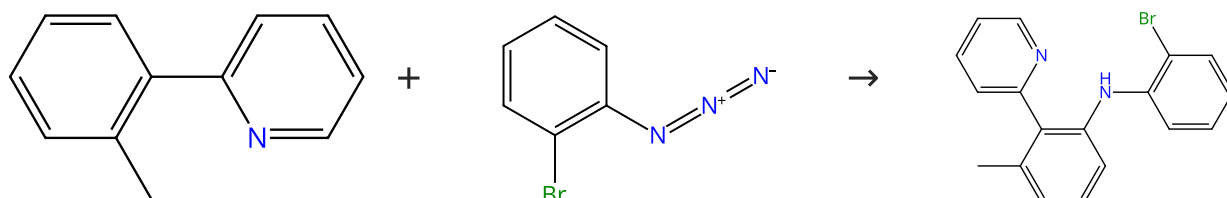
Journal of the American Chemical Society (2013), 135(13), 5000-5003.

1.1 **Reagents:** Pyridine, Tetrabutylammonium tetrafluoroborate, Trifluoromethanesulfonic acid**Solvents:** Acetonitrile; 25 °C1.2 **Reagents:** Piperidine; 12 h, 80 °C

Experimental Protocols

Scheme 40 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (64)

Suppliers (18)

31-080-CAS-98568

Steps: 1 Yield: 98%

Rhodium-Catalyzed Selective Mono- and Diamination of Arenes with Single Directing Site "On Water"

By: Ali, Ashif Md; et al

Organic Letters (2016), 18(6), 1386-1389.

1.1 **Catalysts:** Bis[dichloro[η⁵-(pentamethylcyclopentadienyl)]rhodium], Sodium tetrakis[3,5-bis(trifluoromethyl)phenyl] borate**Solvents:** Water; 24 h, 110 °C; 110 °C → rt

Scheme 41 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (3)

Suppliers (4)

31-080-CAS-3391002

Steps: 1 Yield: 98%

Amination of aminonitropyridine and its N-oxide via vicarious nucleophilic substitution of hydrogen

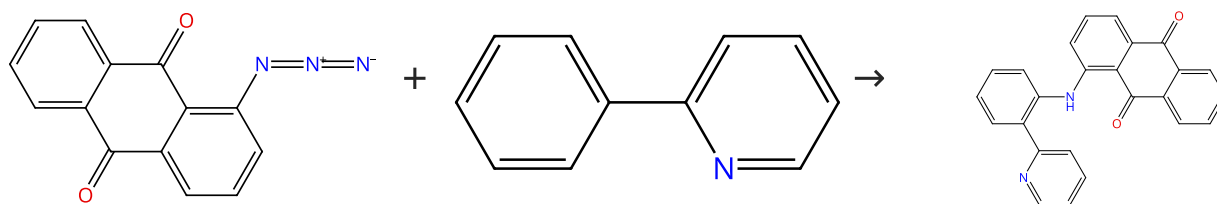
By: Zhao, Xiaofeng; et al

Yingyong Huaxue (2011), 28(4), 376-381.

1.1 **Reagents:** Hydrazinium, 1,1,1-trimethyl-, iodide (1:1)**Solvents:** Dimethyl sulfoxide; rt1.2 **Reagents:** Potassium *tert*-butoxide**Solvents:** Dimethyl sulfoxide; rt; 3 h, rt1.3 **Reagents:** Ammonium chloride**Solvents:** Water

Scheme 42 (1 Reaction)

Steps: 1 Yield: 98%



Supplier (1)

Suppliers (90)

31-080-CAS-9303223

Steps: 1 Yield: 98%

Rhodium-Catalyzed Selective Mono- and Diamination of Arenes with Single Directing Site "On Water"

- 1.1 **Catalysts:** Bis[dichloro[η^5 -(pentamethylcyclopentadienyl)]rhodium], Sodium tetrakis[3,5-bis(trifluoromethyl)phenyl] borate
Solvents: Water; 24 h, 110 °C; 110 °C → rt

By: Ali, Ashif Md; et al

Organic Letters (2016), 18(6), 1386-1389.

Scheme 43 (1 Reaction)

Steps: 1 Yield: 98%



31-080-CAS-1693486

Steps: 1 Yield: 98%

 Rh^{II} -Catalyzed Synthesis of α -, β -, or δ -Carbolines from Aryl Azides

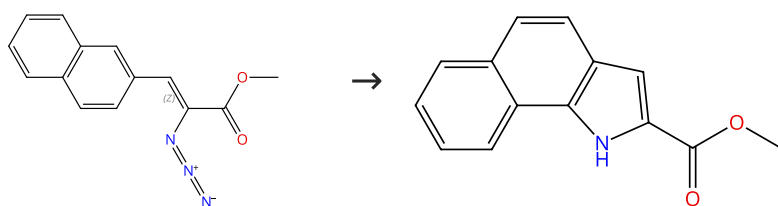
- 1.1 **Catalysts:** Rhodium, bis[μ -[$\alpha,\alpha,\alpha',\alpha'$ -tetramethyl-1,3-benzenedipropanoato(2-)- $\kappa O^1,\kappa O^3:\kappa O^3,\kappa O^1$]]di-, (*Rh-Rh*)
Solvents: 1,2-Dichloroethane; 15 h, 80 °C

By: Pumphrey, Ashley L.; et al

Angewandte Chemie, International Edition (2012), 51(24), 5920-5923, S5920/1-S5920/204.

Scheme 44 (1 Reaction)

Steps: 1 Yield: 98%



Double bond geometry shown

Suppliers (5)

31-080-CAS-6989122

Steps: 1 Yield: 98%

Iron(II) Triflate as a Catalyst for the Synthesis of Indoles by Intramolecular C-H Amination

- 1.1 **Catalysts:** Methanesulfonic acid, 1,1,1-trifluoro-, iron(2+) salt (2:1)
Solvents: Tetrahydrofuran; 24 h, 80 °C

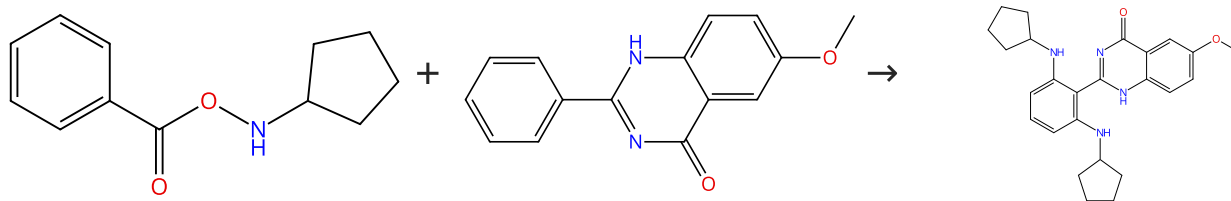
By: Bonnamour, Julien; et al

Organic Letters (2011), 13(8), 2012-2014.

Experimental Protocols

Scheme 45 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (6)

Suppliers (21)

31-080-CAS-19248522

Steps: 1 Yield: 98%

Rhodium(III)-catalyzed C-H amination of 2-arylquinazolin-4(3H)-one with N-alkyl-O-benzoyl-hydroxylamines

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: Pivalic acid, Bis[dichloro[η⁵-(pentamethylcyclopentadienyl)]rhodium]
Solvents: Toluene; 12 h, 50 °C

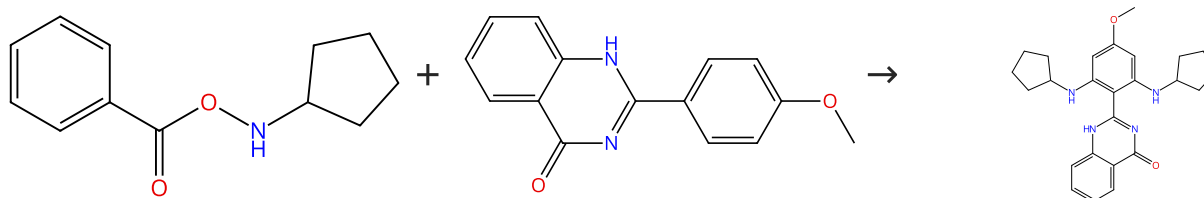
By: Zhang, Yuanguang; et al

Tetrahedron (2018), 74(19), 2330-2337.

Experimental Protocols

Scheme 46 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (6)

Suppliers (37)

31-080-CAS-19248516

Steps: 1 Yield: 98%

Rhodium(III)-catalyzed C-H amination of 2-arylquinazolin-4(3H)-one with N-alkyl-O-benzoyl-hydroxylamines

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: Pivalic acid, Bis[dichloro[η⁵-(pentamethylcyclopentadienyl)]rhodium]
Solvents: Toluene; 12 h, 50 °C

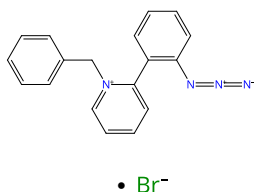
By: Zhang, Yuanguang; et al

Tetrahedron (2018), 74(19), 2330-2337.

Experimental Protocols

Scheme 47 (1 Reaction)

Steps: 1 Yield: 98%



Multi-component
 structure image
 available in CAS
 SciFinder

31-080-CAS-12718346

Steps: 1 Yield: 98%

Rh^{II}-Catalyzed Synthesis of α-, β-, or δ-Carbolines from Aryl Azides

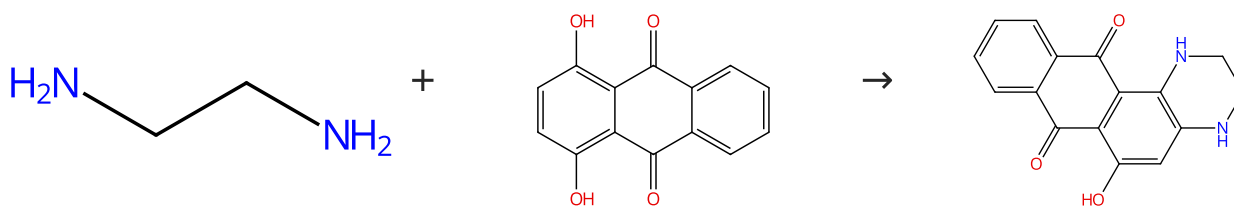
- 1.1 **Catalysts:** Rhodium, bis[μ-[α,α,α',α'-tetramethyl-1,3-benzenedipropanoato(2-)-κO¹,κO³:κO³,κO¹]]di-, (*Rh-Rh*)
Solvents: 1,2-Dichloroethane; 15 h, 80 °C

By: Pumphrey, Ashley L.; et al

Angewandte Chemie, International Edition (2012), 51(24), 5920-5923, S5920/1-S5920/204.

Scheme 48 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (86)

Suppliers (90)

Suppliers (4)

31-032-CAS-11546322

Steps: 1 Yield: 98%

A novel ring-closure reaction between 1,4-dihydroxyanthraquinone and diamines promoted by copper ions

By: Takei, Toshio; et al

Bulletin of the Chemical Society of Japan (1981), 54(9), 2735-8.

1.1 Reagents: Cupric chloride

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